

The Representation Utility Gap: Audio Language Models Encode Clinical Speech Signal Their Answers Do Not Use

Chaitanya Parwatkar, Minoo Ahmadi, Nima Kelidari, Ashutosh Chaubey, Mohammad Soleymani
University of Southern California, Los Angeles, CA, USA

Abstract—Audio language models now answer clinical questions about speech directly: given a recording, a prompt asks whether the speaker is depressed, cognitively impaired, or has Parkinson’s, and the model replies. We test whether those answers use what the models hold. On three conditions chosen so that the words carry nothing, some, or everything of the diagnosis, linear probes read the disease from the frozen encoder of an audio LLM at 0.78 to 0.96 AUC, while the model’s own answers on the same clips sit at 0.54 to 0.66. Transcript-only input lands where audio input lands (0.61 against 0.62 on Alzheimer’s), synthesised speech carrying the same words scores the same as the patient’s voice, and when the words contradict the diagnosis six of nine models fall below chance. Seven fixes that leave the weights untouched, from prompting to activation steering, do not close the gap. Training the smallest parts does: a linear readout on the frozen states reads 0.78 where the answer reads 0.62, and retraining only the projector lifts the model’s own answers to 0.80. The gap replicates on a second Alzheimer’s dataset with new speakers and on the ADReSS-2020 benchmark split. The information is not missing. The readout is.

Index Terms—audio language models, clinical speech, probing, Alzheimer’s disease, Parkinson’s disease, depression

I. Introduction

Speech carries measurable signs of Parkinson’s disease, depression and Alzheimer’s disease, and audio language models (audio LLMs) now answer clinical questions about a recording in natural language. That makes them attractive for screening: one model, one prompt, no task-specific training. Probing studies add to the appeal by showing that speaker state is richly encoded across the layers of these models [1].

The unexamined assumption is that the answer reflects what the layers hold. Clinical evaluations score answers on data where the spoken words and the diagnosis point the same way, so a model that reads the transcript and a model that listens to the voice are indistinguishable by construction. Two recent studies find heavy reliance on text for emotion and synthetic paralinguistic tasks [2], [3]; multi-dataset clinical benchmarks exist but evaluate no audio LLMs, or evaluate text-only models on transcripts [4], [5]. None compares, inside one model, the information present in its representations with the information used by its answer.

We make that comparison on three conditions chosen so that the clinically correct degree of word reliance differs

by construction. In Parkinson’s read speech every speaker reads the same passage, so the words carry nothing. In depression interviews the words carry part of the diagnosis. In Alzheimer’s picture descriptions the words carry most of it. On every condition we train a linear probe on the frozen model’s hidden states and query the same model, on the same clips, with the same question.

Three findings follow. (i) The clinical signal is present: probes read it from the encoder at 0.78 to 0.96 AUC, and on Alzheimer’s it keeps rising through the language model’s own layers. (ii) The answer does not use it: the model’s answers sit at 0.54 to 0.66, transcript-only input lands in the same place, synthesised speech carrying the same words scores the same as the patient’s voice, and when the words contradict the diagnosis the answers invert. (iii) The gap is a readout problem, not a representation problem: seven fixes that leave the weights untouched fail, and the two that train the smallest parts work, on two diseases. Section II gives the design and the controls that keep it honest, Section III the results as a ladder from the gap to the fix, and Section IV what the gap means for evaluating and deploying these models.

II. Setup

Models. Nine audio LLMs, all frozen: qwen2-audio [6], qwen2.5-omni [7], qwen3-omni [8], audio flamingo 2 and 3 [9], [10], kimi-audio [11], salmonn [12], mimo-audio [13] and diva [14]. All nine answer the conflict test; the layer-wise probing and the fixes use qwen2-audio, whose encoder (33 hidden-state stages) and language model (33 stages at the audio-token positions) we probe end to end.

Datasets. Parkinson’s read speech: PC-GITA (Spanish, 1117 clips) [15], Neurovoz (Spanish, 1270) [16], MDVR-KCL (English, 37 speakers) [17]. Depression: DAIC-WOZ interviews (275 participants) [18]. Alzheimer’s: the DementiaBank Pitt cookie-theft descriptions (468 thirty-second segments, 228 speakers) [19], [20], the ADReSSo recordings (237 speakers) [21], and the ADReSS-2020 benchmark split (108 train, 48 test) [22]. An Italian Parkinson’s corpus was excluded after recording properties alone separated its groups at 0.995 (Section II-A).

Prompt and metric. Verbatim, with no transcript supplied in any audio condition: “Based only on this recording, does this speaker show signs of X? Answer with one word,

Yes or No.” We read $p_{\text{yes}} = P(\text{Yes})/(P(\text{Yes}) + P(\text{No}))$ from the logits at the first answer position, one forward pass, no generation, and report AUC against the clinical label. We also record the mass $P(\text{Yes}) + P(\text{No})$ and the yes-rate, because some models answer Yes on every clip (qwen2-audio on PC-GITA) or No on every clip (qwen2.5-omni); those AUCs rank confidences that never cross the threshold, and we flag them.

Probe. An ℓ_2 -regularised logistic probe on the mean-pooled hidden state of one layer, with standardisation fitted inside the training fold. Folds are speaker-disjoint (grouped 5-fold). Where a layer is chosen, it is chosen inside the training folds; selecting the layer after seeing the test fold inflates AUC by +0.03 on average across 18 cells and by up to +0.14, so we quote the nested value wherever both exist.

Conflict sets. From the raw DAIC transcripts we take 26,083 participant turns, score each with a text sentiment model, and keep the moments where a depressed participant speaks positively (138) or a control speaks negatively (57), paired with matched agreement segments from the same speakers: 195 pairs over 74 speakers, with label and lexical direction independent ($\phi = 0.000$). On Pitt we score the words of each segment with a content-marker fluency model, after removing speaking rate, which alone reaches 0.92 and would make the split trivial; 146 segments land in conflict and 322 in agreement. Nothing is synthesised: the voices are real patients and the contradicting words occur naturally.

Transcript and synthesis controls. For the transcript-only condition the model receives the segment’s transcript (official for Pitt, Whisper [23] for ADReSSo) with the same question and no audio. For the synthesis condition each transcript is spoken by a neutral Piper voice [24] and scored like the real recording.

A. Controls on the measurement

Before any result depends on the probe, we checked what else it could be reading. Six recording measurements containing no speech content separate patients from controls at 0.995 on the Italian corpus, and two groups of healthy elderly controls from that cohort recorded on different days separate at 1.000; we exclude it. On the four Parkinson’s and depression corpora the recording-only floor is 0.45 to 0.80 and the probe stays above it; on Neurovoz, age matching moves the encoder probe from 0.95 to 0.91 while a silence-only probe falls from 0.72 to 0.58. On Pitt the strongest channel feature, duration, reaches 0.61 and all six together 0.65, below the probe. Finally the signal travels: a probe trained on PC-GITA read speech and tested cold on MDVR-KCL, a different language and recording setup, reaches 0.86 (Table III).

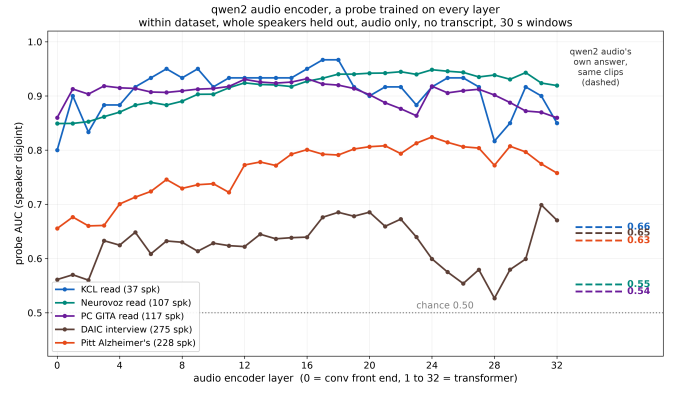


Fig. 1. A linear probe on every encoder layer of qwen2-audio (solid) against the model’s own answer on the same clips (dashed, one line per task). On the three Parkinson’s sets the probe sits at 0.89 to 0.96 while the answers sit at 0.54 to 0.66; on Alzheimer’s the probe reaches 0.78 under strict speaker-disjoint selection while the answer reads 0.62. On depression at a 30s window both are near 0.66, and that condition’s evidence comes from the conflict test instead.

III. Results

A. The signal is there, and the answer does not use it

Figure 1 shows the gap directly. Under nested layer selection the encoder probe reads 0.92 on PC-GITA, 0.96 on Neurovoz and 0.89 on MDVR-KCL read speech, and 0.78 on the Pitt Alzheimer’s segments (0.82 at the curve’s peak); the same model’s answers on the same clips read 0.54, 0.55, 0.66 and 0.62. On Alzheimer’s the signal does not stop at the encoder: probing the language model’s own stages at the audio-token positions, the readable signal rises to 0.86 by the deep layers while the answer stays where it was. On depression at the 30s window qwen2-audio accepts, the nested probe (0.66) and the answer (0.65) coincide; the depression evidence below rests on the conflict construction, not on this gap.

B. What the answer uses instead: the words

Three controls agree. Transcript only. Given the Pitt transcript and no audio, qwen2-audio scores 0.61 (95% bootstrap interval 0.56 to 0.67); with the audio it scores 0.62 (0.57 to 0.67). On ADReSSo the pair is 0.62 and 0.64. On DAIC the transcript-only answer of qwen2.5-omni correlates with its own audio answer at $r = 0.83$ and returns the identical Yes/No on 96% of segments, and across the depression grid no model beats its own transcript (0 of 6; median text 0.81, audio 0.83). Synthesised voice. Replacing each Pitt patient’s voice with a neutral synthetic voice speaking the same words leaves the answers essentially unchanged (qwen2-audio 0.62 real, 0.60 synthetic; qwen2.5-omni 0.67 synthetic; diva 0.69; mimo-audio 0.58) and the transcript alone does best of all (0.72, 0.72, 0.71, 0.57). The same shape holds on DAIC. Words check. A probe trained only on Pitt segments where the words agree with the diagnosis reaches 0.95 within that set and at most 0.58 on the segments where the words

TABLE I

Depression conflict test, 195 matched pairs, no transcript given. AUC against the clinical label; below 0.5 means the model follows the words rather than the diagnosis. Six of nine models invert. Audio flamingo 2 is flat on both arms and has no signal to invert.

model	conflict	agreement
salmonn	0.29	0.85
diva	0.30	0.70
qwen2.5-omni	0.31	0.85
qwen3-omni	0.35	0.87
qwen2-audio	0.35	0.92
audio flamingo 3	0.36	0.93
mimo-audio	0.47	0.82
kimi-audio	0.52	0.87
audio flamingo 2	0.55	0.57

TABLE II

The answer on the conflict segments under each fix (AUC). The first block leaves the weights untouched, on the depression conflict pairs; the second, on the Alzheimer’s conflict segments. Only the two fixes that train something move it.

fix	conflict AUC
depression, no training	
the model’s own answer	0.30
best of 12 prompt wordings	0.48
text subtraction, best weight	0.50
activation steering, best strength	0.54
Alzheimer’s, no training	
the model’s own answer	0.41
few-shot, example audios in the prompt	0.46
probe’s best layer into the projector, untrained	0.55
Alzheimer’s, a small part trained	
linear readout on the frozen states	0.61
projector retrained on layer 24	0.57

contradict it, with no speaker shared; on ADReSSo the same test gives 0.99 within and 0.39 to 0.61 on conflict, on a small conflict group of 19. What the probe learns on this task is, to a large degree, the words, and they enter at the encoder itself, which was trained to transcribe.

C. The stress test: when words and diagnosis disagree

Table I gives the depression conflict test. Six of nine models, from five independent labs, score below chance where the words point away from the diagnosis, as low as 0.29, while scoring 0.70 to 0.93 on the matched segments where words and diagnosis agree. On the Pitt Alzheimer’s conflict segments qwen2-audio reads 0.41 against 0.70 on agreement. A model that listened would not invert; a model that reads must.

D. Fixes that leave the weights alone fail

Table II lists the attempts. Twelve prompt wordings never lift the depression conflict AUC above 0.48. Learned layer mixing over the encoder reaches 0.909 against 0.916 for choosing one layer honestly, and conditioning the mixture on the prompt adds 0.0004. Subtracting the transcript-only logits from the audio logits at weight α

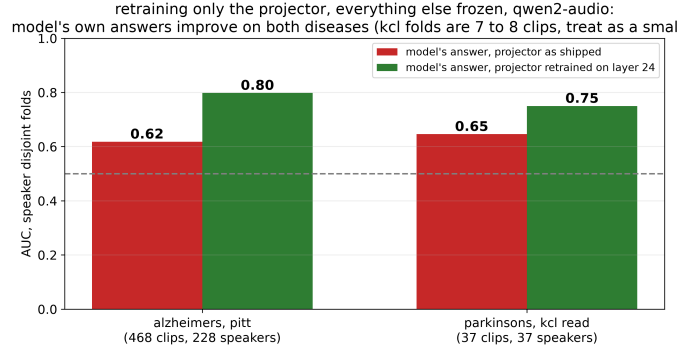


Fig. 2. Retraining only the projector, everything else frozen, speaker-disjoint folds. The model’s own answers improve on both diseases. On Alzheimer’s the gain is +0.18 (95% interval 0.12 to 0.24); on MDVR-KCL, with 37 speakers, it is +0.11 (interval −0.12 to 0.35), a direction rather than a significant result.

makes the model stop following the words at $\alpha = 0.9$ (agreement with the lexical direction falls from 0.70 to 0.50) but conflict AUC only reaches 0.50, and on held-out speakers $\alpha = 0$ is chosen in four of five folds. Adding a multiple of the difference-in-means diagnosis direction to the encoder output damages the agreement arm at the smallest coefficient before conflict moves, and a random direction of the same norm displaces the answer equally; across 20 steered conditions the apparent gain correlates at -0.92 with the lexical signal being destroyed. Placing example recordings in the prompt collapses the answers to a single class. Feeding the probe’s best encoder layer into the projector without training breaks the answer (0.62 to 0.49 overall): the projector wants the layer it was trained on.

E. Training the smallest part works

Two interventions train a small component and both close most of the gap. A linear readout on the frozen hidden states, which is the probe itself, reads 0.78 (interval 0.74 to 0.82) on the Pitt segments where the answer reads 0.62, a paired gain of +0.17 (0.10 to 0.23). Retraining only the projector, the connector between the audio encoder and the language model, on the diagnosis answer, with the encoder and the language model frozen and encoder layer 24 as its input, lifts the model’s own answers from 0.62 to 0.80 (0.76 to 0.84), a paired gain of +0.18 (0.12 to 0.24), and from 0.70 to 0.89 on the agreement segments. On the conflict segments the readout reaches 0.61 and the retrained projector 0.57: word conflict remains the hard case, and only the readout partly rescues it. The same projector recipe on MDVR-KCL moves the Parkinson’s answers from 0.65 to 0.75 (Figure 2).

F. The gap is not one dataset’s

On ADReSSo, 237 speakers none of whom appear in Pitt, the probe reaches 0.90 and the model’s answer

TABLE III

A probe trained on one Parkinson’s read-speech dataset and tested cold on another (qwen2-audio encoder). The diagonal is within-dataset. Transfer is partial but real across language and recording setup.

trained on \ tested on	KCL (en)	Neurovoz (es)	PC-GITA (es)
KCL	0.97	0.58	0.64
Neurovoz	0.74	0.95	0.66
PC-GITA	0.86	0.46	0.93

0.64, with transcript-only input at 0.62. On the ADReSS-2020 benchmark split, with the layer chosen inside the 108 training recordings, the probe reads the 48 test recordings at AUC 0.78 and accuracy 71%, against the challenge’s own acoustic baseline of 62.5%; pooled over all 156 recordings the probe reads 0.82 and the answer 0.65. Across the Parkinson’s sets the probe transfers across language (Table III).

IV. Discussion

Read together, the ladder has one shape. The clinical signal is present at every depth of the model, the answer does not use it, and what the answer uses instead is the words. Everything that leaves the weights untouched fails to change this, and everything that trains the smallest part succeeds, on two diseases. The deficit therefore lives in the readout, not in the representation: these models are better clinical encoders than clinical reporters, and evaluations that score their answers only where words and diagnosis agree cannot tell a model that listens from one that reads. Limitations. The layer-wise probing and the fixes are on one model; the nine-model evidence is the answer-side conflict test. The depression probe and answer coincide at the 30s window, so that condition’s claim rests on the conflict construction. MDVR-KCL has 37 speakers and its projector gain is not significant on its own. ADReSSo transcripts are ASR output, its conflict group has 19 segments, and its ADReSS-2020 test labels were recovered by matching transcripts to Pitt, a procedure that scored 107 of 108 with no errors on the training recordings whose labels are known. The encoder association, that 6 of 7 transcription-trained encoders invert against 0 of 2 that were not (Fisher $p = 0.08$), is suggestive and post hoc. On DAIC, scores track PTSD severity as closely as depression severity (0.60 against 0.59), so the models detect distress rather than depression specifically. Pitt and DAIC are distributed under their own agreements; we release segment identifiers, scripts and all per-clip scores, not audio or text.

References

- [1] Y. Belinkov, “Probing classifiers: Promises, shortcomings, and advances,” *Computational Linguistics*, vol. 48, no. 1, pp. 207–219, 2022. [Online]. Available: <https://aclanthology.org/2022.cl-1.7/>
- [2] J. Chen, Z. Guo, J. Chun, P. Wang, A. Perrault, and M. Elsner, “Do audio LLMs really LISTEN, or just transcribe? measuring lexical vs. acoustic emotion cues reliance,” in *Proceedings of the 19th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)*. Rabat, Morocco: Association for Computational Linguistics, 2026, pp. 5848–5877. [Online]. Available: <https://aclanthology.org/2026.eacl-long.274/>
- [3] J. Pang, A. Chaubey, and M. Soleymani, “Do audio LLMs listen or read? analyzing and mitigating paralinguistic failures with VoxParadox,” in *Proceedings of the 43rd International Conference on Machine Learning (ICML)*, 2026. [Online]. Available: <https://arxiv.org/abs/2605.27772>
- [4] S. Bhalla, L. Kieu, A. Merchant, E. de Lara, and A. Mariakakis, “SpeechDx: A multi-task benchmark for clinical speech AI,” *arXiv preprint arXiv:2606.17339*, 2026. [Online]. Available: <https://arxiv.org/abs/2606.17339>
- [5] Z. Dong, H. Guo, W. Xu, H. Zhao, and Z. Zhang, “Foundation model-based evaluation of neuropsychiatric disorders: A lifespan-inclusive, multi-modal, and multi-lingual study,” *arXiv preprint arXiv:2512.20948*, 2025. [Online]. Available: <https://arxiv.org/abs/2512.20948>
- [6] Y. Chu, J. Xu, Q. Yang, H. Wei, X. Wei, Z. Guo, Y. Leng, Y. Lv, J. He, J. Lin, C. Zhou, and J. Zhou, “Qwen2-audio technical report,” *arXiv preprint arXiv:2407.10759*, 2024. [Online]. Available: <https://arxiv.org/abs/2407.10759>
- [7] J. Xu, Z. Guo, J. He, H. Hu, T. He, S. Bai, K. Chen, J. Wang, Y. Fan, K. Dang, B. Zhang, X. Wang, Y. Chu, and J. Lin, “Qwen2.5-omni technical report,” *arXiv preprint arXiv:2503.20215*, 2025. [Online]. Available: <https://arxiv.org/abs/2503.20215>
- [8] J. Xu, Z. Guo, H. Hu, Y. Chu, X. Wang, J. He, Y. Wang, X. Shi, T. He, X. Zhu, Y. Lv, Y. Wang, D. Guo, H. Wang, L. Ma, P. Zhang, X. Zhang, H. Hao, Z. Guo, B. Yang, B. Zhang, Z. Ma, X. Wei, S. Bai, K. Chen, X. Liu, P. Wang, M. Yang, D. Liu, X. Ren, B. Zheng, R. Men, F. Zhou, B. Yu, J. Yang, L. Yu, J. Zhou, and J. Lin, “Qwen3-omni technical report,” *arXiv preprint arXiv:2509.17765*, 2025. [Online]. Available: <https://arxiv.org/abs/2509.17765>
- [9] S. Ghosh, Z. Kong, S. Kumar, S. Sakshi, J. Kim, W. Ping, R. Valle, D. Manocha, and B. Catanzaro, “Audio flamingo 2: An audio-language model with long-audio understanding and expert reasoning abilities,” in *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, ser. *Proceedings of Machine Learning Research*, vol. 267. PMLR, 2025, pp. 19358–19405. [Online]. Available: <https://proceedings.mlr.press/v267/ghosh25b.html>
- [10] A. Goel, S. Ghosh, J. Kim, S. Kumar, Z. Kong, S. Gil Lee, C.-H. H. Yang, R. Duraiswami, D. Manocha, R. Valle, and B. Catanzaro, “Audio flamingo 3: Advancing audio intelligence with fully open large audio language models,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2025. [Online]. Available: <https://arxiv.org/abs/2507.08128>
- [11] Kimi Team, D. Ding, Z. Ju, Y. Leng, S. Liu, T. Liu, Z. Shang, K. Shen, W. Song, X. Tan, H. Tang, Z. Wang, C. Wei, Y. Xin, X. Xu, J. Yu, Y. Zhang, X. Zhou, Y. Charles, J. Chen, Y. Chen, Y. Du, W. He, Z. Hu, G. Lai, Q. Li, Y. Liu, W. Sun, J. Wang, Y. Wang, Y. Wu, Y. Wu, D. Yang, H. Yang, Y. Yang, Z. Yang, A. Yin, R. Yuan, Y. Zhang, and Z. Zhou, “Kimi-audio technical report,” *arXiv preprint arXiv:2504.18425*, 2025. [Online]. Available: <https://arxiv.org/abs/2504.18425>
- [12] C. Tang, W. Yu, G. Sun, X. Chen, T. Tan, W. Li, L. Lu, Z. Ma, and C. Zhang, “SALMONN: Towards generic hearing abilities for large language models,” in *The Twelfth International Conference on Learning Representations (ICLR)*, 2024. [Online]. Available: <https://openreview.net/forum?id=14rn7HpKVk>
- [13] Xiaomi LLM-Core Team, “MiMo-Audio: Audio language models are few-shot learners,” *arXiv preprint arXiv:2512.23808*, 2025, full 100-person author list on arXiv. [Online]. Available: <https://arxiv.org/abs/2512.23808>
- [14] W. Held, Y. Zhang, M. Li, W. Shi, M. J. Ryan, and D. Yang, “Distilling an end-to-end voice assistant without instruction training data,” in *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics*

- (Volume 1: Long Papers). Vienna, Austria: Association for Computational Linguistics, 2025, pp. 7876–7891. [Online]. Available: <https://aclanthology.org/2025.acl-long.388/>
- [15] J. R. Orozco-Arroyave, J. D. Arias-Londoño, J. F. Vargas-Bonilla, M. C. González-Rátiva, and E. Nöth, “New Spanish speech corpus database for the analysis of people suffering from Parkinson’s disease,” in Proceedings of the Ninth International Conference on Language Resources and Evaluation (LREC’14). Reykjavik, Iceland: European Language Resources Association (ELRA), 2014, pp. 342–347. [Online]. Available: <https://aclanthology.org/L14-1549/>
 - [16] J. Mendes-Laureano, J. A. Gómez-García, A. Guerrero-López, E. Luque-Buzo, J. D. Arias-Londoño, F. J. Grandas-Pérez, and J. I. Godino-Llorente, “NeuroVoz: A Castillian Spanish corpus of parkinsonian speech,” *Scientific Data*, vol. 11, no. 1, p. 1367, 2024.
 - [17] H. Jaeger, D. Trivedi, and M. Stadtschnitzer, “Mobile device voice recordings at King’s college London (MDVR-KCL) from both early and advanced Parkinson’s disease patients and healthy controls,” Zenodo, 2019. [Online]. Available: <https://doi.org/10.5281/zenodo.2867216>
 - [18] J. Gratch, R. Artstein, G. Lucas, G. Stratou, S. Scherer, A. Nazarian, R. Wood, J. Boberg, D. DeVault, S. Marsella, D. Traum, S. Rizzo, and L.-P. Morency, “The distress analysis interview corpus of human and computer interviews,” in Proceedings of the Ninth International Conference on Language Resources and Evaluation (LREC’14). Reykjavik, Iceland: European Language Resources Association (ELRA), 2014, pp. 3123–3128. [Online]. Available: <https://aclanthology.org/L14-1421/>
 - [19] J. T. Becker, F. Boller, O. L. Lopez, J. Saxton, and K. L. McGonigle, “The natural history of Alzheimer’s disease: Description of study cohort and accuracy of diagnosis,” *Archives of Neurology*, vol. 51, no. 6, pp. 585–594, 1994.
 - [20] A. M. Lanzi, A. K. Saylor, D. Fromm, H. Liu, B. MacWhinney, and M. L. Cohen, “DementiaBank: Theoretical rationale, protocol, and illustrative analyses,” *American Journal of Speech-Language Pathology*, vol. 32, no. 2, pp. 426–438, 2023.
 - [21] S. Luz, F. Haider, S. de la Fuente, D. Fromm, and B. MacWhinney, “Detecting cognitive decline using speech only: The ADReSSo challenge,” in *Proc. Interspeech 2021*, 2021, pp. 3780–3784.
 - [22] —, “Alzheimer’s dementia recognition through spontaneous speech: The ADReSS challenge,” in *Proc. Interspeech 2020*, 2020, pp. 2172–2176.
 - [23] A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey, and I. Sutskever, “Robust speech recognition via large-scale weak supervision,” in Proceedings of the 40th International Conference on Machine Learning (ICML), ser. Proceedings of Machine Learning Research, vol. 202. PMLR, 2023, pp. 28 492–28 518. [Online]. Available: <https://proceedings.mlr.press/v202/radford23a.html>
 - [24] M. Hansen, “Piper: A fast, local neural text to speech system,” GitHub repository, <https://github.com/rhasspy/piper>, 2023, mIT license; repository archived October 2025, development continued at OHF-Voice/piper1-gpl.